## Graphs of Functions

If $f$ is a function with domain $D$, its **graph** consists of the points in the Cartesian plane whose coordinates are the input-output pairs for $f$. In set notation, the graph is

$$\{(x, f(x)) \mid x \in D\}.$$

The graph of the function $f(x) = x + 2$ is the set of points with coordinates $(x, y)$ for which $y = x + 2$. Its graph is the straight line sketched in Figure 1.3.

The graph of a function $f$ is a useful picture of its behavior. If $(x, y)$ is a point on the graph, then $y = f(x)$ is the height of the graph above (or below) the point $x$. The height may be positive or negative, depending on the sign of $f(x)$ (Figure 1.4).

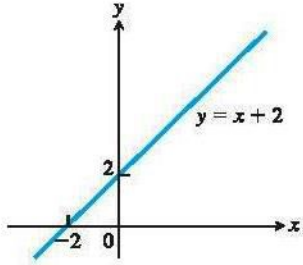



**FIGURE 1.3** The graph of $f(x) = x + 2$ is the set of points $(x, y)$ for which $y$ has the value $x + 2$.

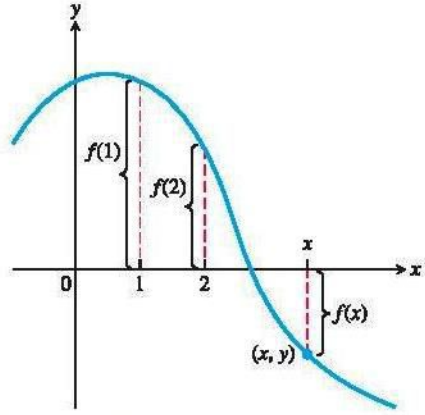



**FIGURE 1.4** If $(x, y)$ lies on the graph of $f$, then the value $y = f(x)$ is the height of the graph above the point $x$ (or below $x$ if $f(x)$ is negative).

| $x$ | $y = x^2$ |
|---|---|
| $-2$ | 4 |
| $-1$ | 1 |
| 0 | 0 |
| 1 | 1 |
| $\frac{3}{2}$ | $\frac{9}{4}$ |
| 2 | 4 |

**EXAMPLE 2** Graph the function $y = x^2$ over the interval $[-2, 2]$.

**Solution** Make a table of $xy$-pairs that satisfy the equation $y = x^2$. Plot the points $(x, y)$ whose coordinates appear in the table, and draw a *smooth* curve (labeled with its equation) through the plotted points (see Figure 1.5). ■

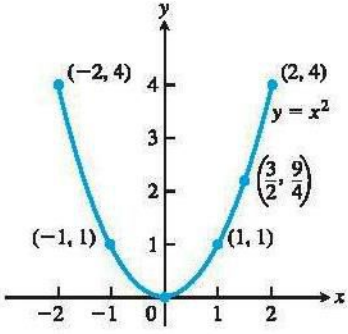
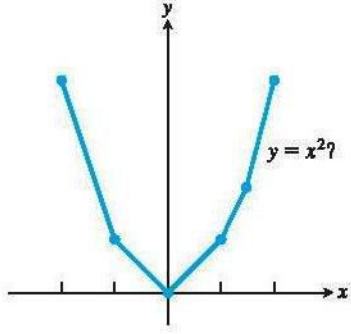
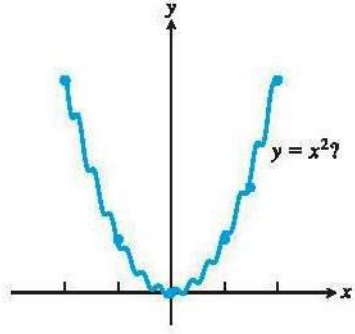



**FIGURE 1.5** Graph of the function in Example 2.

How do we know that the graph of $y = x^2$ doesn't look like one of these curves?



To find out, we could plot more points. But how would we then connect *them*? The basic question still remains: How do we know for sure what the graph looks like between the points we plot? Calculus answers this question, as we will see in Chapter 4. Meanwhile, we will have to settle for plotting points and connecting them as best we can.

## Representing a Function Numerically

We have seen how a function may be represented algebraically by a formula (the area function) and visually by a graph (Example 2). Another way to represent a function is **numerically**, through a table of values. Numerical representations are often used by engineers and experimental scientists. From an appropriate table of values, a graph of the function can be obtained using the method illustrated in Example 2, possibly with the aid of a computer. The graph consisting of only the points in the table is called a **scatterplot**.

**EXAMPLE 3** Musical notes are pressure waves in the air. The data associated with Figure 1.6 give recorded pressure displacement versus time in seconds of a musical note produced by a tuning fork. The table provides a representation of the pressure function over time. If we first make a scatterplot and then connect approximately the data points $(t, p)$ from the table, we obtain the graph shown in the figure.

| Time | Pressure | Time | Pressure |
|---|---|---|---|
| 0.00091 | −0.080 | 0.00362 | 0.217 |
| 0.00108 | 0.200 | 0.00379 | 0.480 |
| 0.00125 | 0.480 | 0.00398 | 0.681 |
| 0.00144 | 0.693 | 0.00416 | 0.810 |
| 0.00162 | 0.816 | 0.00435 | 0.827 |
| 0.00180 | 0.844 | 0.00453 | 0.749 |
| 0.00198 | 0.771 | 0.00471 | 0.581 |
| 0.00216 | 0.603 | 0.00489 | 0.346 |
| 0.00234 | 0.368 | 0.00507 | 0.077 |
| 0.00253 | 0.099 | 0.00525 | −0.164 |
| 0.00271 | −0.141 | 0.00543 | −0.320 |
| 0.00289 | −0.309 | 0.00562 | −0.354 |
| 0.00307 | −0.348 | 0.00579 | −0.248 |
| 0.00325 | −0.248 | 0.00598 | −0.035 |
| 0.00344 | −0.041 | | |

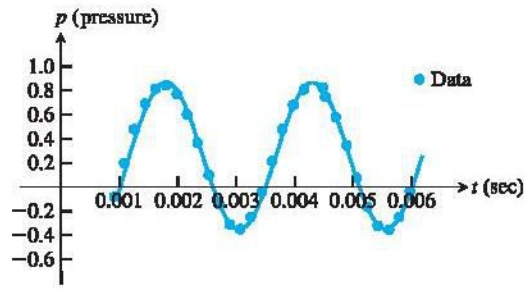



**FIGURE 1.6** A smooth curve through the plotted points gives a graph of the pressure function represented by the accompanying tabled data (Example 3).

## The Vertical Line Test for a Function

Not every curve in the coordinate plane can be the graph of a function. A function $f$ can have only one value $f(x)$ for each $x$ in its domain, so *no vertical* line can intersect the graph of a function more than once. If $a$ is in the domain of the function $f$, then the vertical line $x = a$ will intersect the graph of $f$ at the single point $(a, f(a))$.

A circle cannot be the graph of a function, since some vertical lines intersect the circle twice. The circle graphed in Figure 1.7a, however, does contain the graphs of functions of $x$, such as the upper semicircle defined by the function $f(x) = \sqrt{1 - x^2}$ and the lower semicircle defined by the function $g(x) = -\sqrt{1 - x^2}$ (Figures 1.7b and 1.7c).

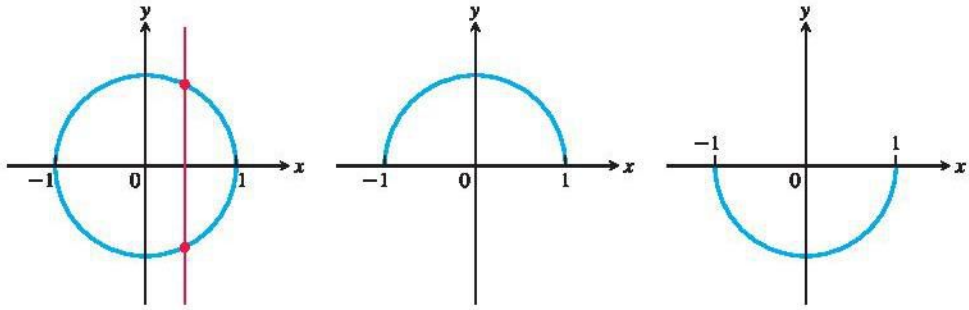



**FIGURE 1.7** (a) The circle is not the graph of a function; it fails the vertical line test. (b) The upper semicircle is the graph of a function $f(x) = \sqrt{1 - x^2}$. (c) The lower semicircle is the graph of a function $g(x) = -\sqrt{1 - x^2}$.

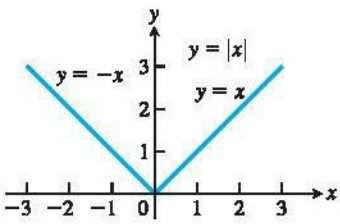



**FIGURE 1.8** The absolute value function has domain $(-\infty, \infty)$ and range $[0, \infty)$.

## Piecewise-Defined Functions

Sometimes a function is described in pieces by using different formulas on different parts of its domain. One example is the **absolute value function**

$$|x| = \begin{cases} x, & x \geq 0 \quad \text{First formula} \\ -x, & x < 0, \quad \text{Second formula} \end{cases}$$

whose graph is given in Figure 1.8. The right-hand side of the equation means that the function equals $x$ if $x \geq 0$, and equals $-x$ if $x < 0$. Piecewise-defined functions often arise when real-world data are modeled. Here are some other examples.

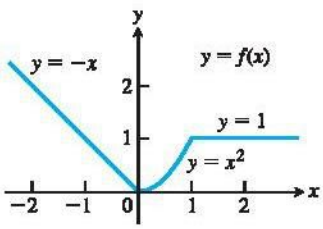



**FIGURE 1.9** To graph the function $y = f(x)$ shown here, we apply different formulas to different parts of its domain (Example 4).

**EXAMPLE 4** The function

$$f(x) = \begin{cases} -x, & x < 0 \quad \text{First formula} \\ x^2, & 0 \leq x \leq 1 \quad \text{Second formula} \\ 1, & x > 1 \quad \text{Third formula} \end{cases}$$

is defined on the entire real line but has values given by different formulas, depending on the position of $x$. The values of $f$ are given by $y = -x$ when $x < 0$, $y = x^2$ when $0 \leq x \leq 1$, and $y = 1$ when $x > 1$. The function, however, is *just one function* whose domain is the entire set of real numbers (Figure 1.9). ■

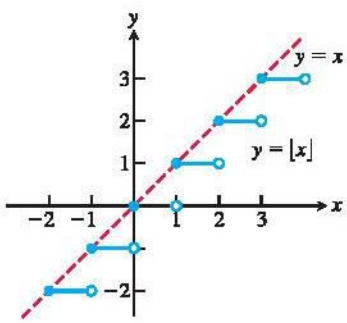



**FIGURE 1.10** The graph of the greatest integer function $y = \lfloor x \rfloor$ lies on or below the line $y = x$, so it provides an integer floor for $x$ (Example 5).

**EXAMPLE 5** The function whose value at any number $x$ is the *greatest integer less than or equal to x* is called the **greatest integer function** or the **integer floor function**. It is denoted $\lfloor x \rfloor$. Figure 1.10 shows the graph. Observe that

$$\lfloor 2.4 \rfloor = 2, \quad \lfloor 1.9 \rfloor = 1, \quad \lfloor 0 \rfloor = 0, \quad \lfloor -1.2 \rfloor = -2,$$
$$\lfloor 2 \rfloor = 2, \quad \lfloor 0.2 \rfloor = 0, \quad \lfloor -0.3 \rfloor = -1, \quad \lfloor -2 \rfloor = -2.$$ ■

**EXAMPLE 6** The function whose value at any number $x$ is the *smallest integer greater than or equal to x* is called the **least integer function** or the **integer ceiling function**. It is denoted $\lceil x \rceil$. Figure 1.11 shows the graph. For positive values of $x$, this function might represent, for example, the cost of parking $x$ hours in a parking lot that charges \$1 for each hour or part of an hour. ■

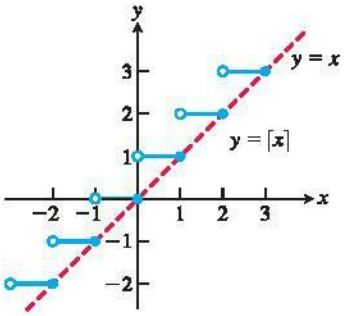



**FIGURE 1.11** The graph of the least integer function $y = \lceil x \rceil$ lies on or above the line $y = x$, so it provides an integer ceiling for $x$ (Example 6).

## Increasing and Decreasing Functions

If the graph of a function *climbs* or *rises* as you move from left to right, we say that the function is *increasing*. If the graph *descends* or *falls* as you move from left to right, the function is *decreasing*.

> **DEFINITIONS** Let $f$ be a function defined on an interval $I$ and let $x_1$ and $x_2$ be any two points in $I$.
>
> 1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then $f$ is said to be **increasing** on $I$.
> 2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then $f$ is said to be **decreasing** on $I$.

It is important to realize that the definitions of increasing and decreasing functions must be satisfied for *every* pair of points $x_1$ and $x_2$ in $I$ with $x_1 < x_2$. Because we use the inequality $<$ to compare the function values, instead of $\leq$, it is sometimes said that $f$ is *strictly* increasing or decreasing on $I$. The interval $I$ may be finite (also called bounded) or infinite (unbounded) and by definition never consists of a single point (Appendix 1).

**EXAMPLE 7** The function graphed in Figure 1.9 is decreasing on $(-\infty, 0]$ and increasing on $[0, 1]$. The function is neither increasing nor decreasing on the interval $[1, \infty)$ because of the strict inequalities used to compare the function values in the definitions. ■

## Even Functions and Odd Functions: Symmetry

The graphs of *even* and *odd* functions have characteristic symmetry properties.

> **DEFINITIONS** A function $y = f(x)$ is an
>
> **even function of $x$** if $f(-x) = f(x)$,
> **odd function of $x$** if $f(-x) = -f(x)$,
>
> for every $x$ in the function's domain.

The names *even* and *odd* come from powers of $x$. If $y$ is an even power of $x$, as in $y = x^2$ or $y = x^4$, it is an even function of $x$ because $(-x)^2 = x^2$ and $(-x)^4 = x^4$. If $y$ is an odd power of $x$, as in $y = x$ or $y = x^3$, it is an odd function of $x$ because $(-x)^1 = -x$ and $(-x)^3 = -x^3$.

The graph of an even function is **symmetric about the $y$-axis**. Since $f(-x) = f(x)$, a point $(x, y)$ lies on the graph if and only if the point $(-x, y)$ lies on the graph (Figure 1.12a). A reflection across the $y$-axis leaves the graph unchanged.

The graph of an odd function is **symmetric about the origin**. Since $f(-x) = -f(x)$, a point $(x, y)$ lies on the graph if and only if the point $(-x, -y)$ lies on the graph (Figure 1.12b). Equivalently, a graph is symmetric about the origin if a rotation of 180° about the origin leaves the graph unchanged. Notice that the definitions imply that both $x$ and $-x$ must be in the domain of $f$.

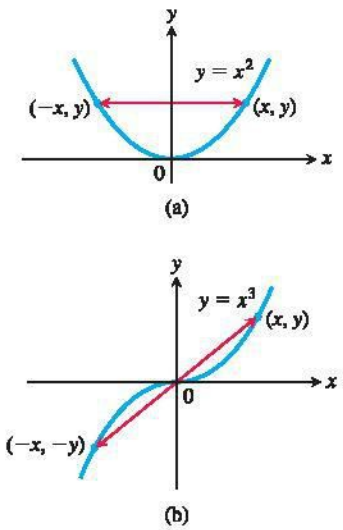



**FIGURE 1.12** (a) The graph of $y = x^2$ (an even function) is symmetric about the $y$-axis. (b) The graph of $y = x^3$ (an odd function) is symmetric about the origin.

**EXAMPLE 8** Here are several functions illustrating the definition.

| | |
|---|---|
| $f(x) = x^2$ | Even function: $(-x)^2 = x^2$ for all $x$; symmetry about $y$-axis. |
| $f(x) = x^2 + 1$ | Even function: $(-x)^2 + 1 = x^2 + 1$ for all $x$; symmetry about $y$-axis (Figure 1.13a). |
| $f(x) = x$ | Odd function: $(-x) = -x$ for all $x$; symmetry about the origin. |
| $f(x) = x + 1$ | Not odd: $f(-x) = -x + 1$, but $-f(x) = -x - 1$. The two are not equal.<br>Not even: $(-x) + 1 \neq x + 1$ for all $x \neq 0$ (Figure 1.13b). ■ |

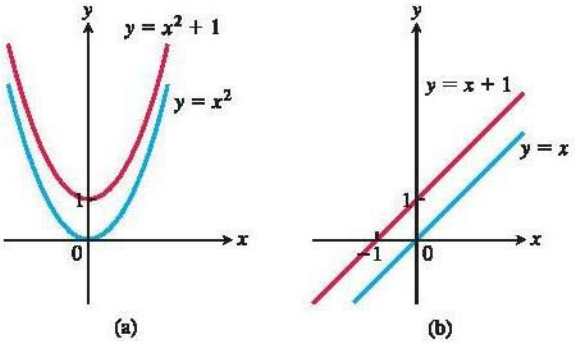



**FIGURE 1.13** (a) When we add the constant term 1 to the function $y = x^2$, the resulting function $y = x^2 + 1$ is still even and its graph is still symmetric about the $y$-axis. (b) When we add the constant term 1 to the function $y = x$, the resulting function $y = x + 1$ is no longer odd, since the symmetry about the origin is lost. The function $y = x + 1$ is also not even (Example 8).

## Common Functions

A variety of important types of functions are frequently encountered in calculus. We identify and briefly describe them here.

**Linear Functions** A function of the form $f(x) = mx + b$, for constants $m$ and $b$, is called a **linear function**. Figure 1.14a shows an array of lines $f(x) = mx$ where $b = 0$, so these lines pass through the origin. The function $f(x) = x$ where $m = 1$ and $b = 0$ is called the **identity function**. Constant functions result when the slope $m = 0$ (Figure 1.14b). A linear function with positive slope whose graph passes through the origin is called a *proportionality* relationship.

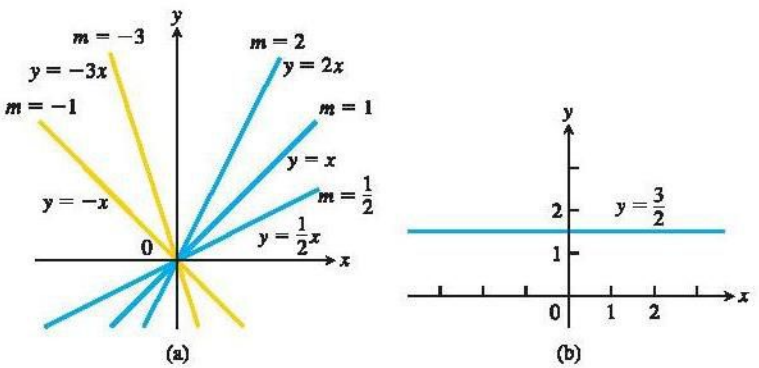



**FIGURE 1.14** (a) Lines through the origin with slope $m$. (b) A constant function with slope $m = 0$.

> **DEFINITION** Two variables $y$ and $x$ are **proportional** (to one another) if one is always a constant multiple of the other; that is, if $y = kx$ for some nonzero constant $k$.

If the variable $y$ is proportional to the reciprocal $1/x$, then sometimes it is said that $y$ is **inversely proportional** to $x$ (because $1/x$ is the multiplicative inverse of $x$).

**Power Functions** A function $f(x) = x^a$, where $a$ is a constant, is called a **power function**. There are several important cases to consider.

**(a)** $a = n$, a positive integer.

The graphs of $f(x) = x^n$, for $n = 1, 2, 3, 4, 5$, are displayed in Figure 1.15. These functions are defined for all real values of $x$. Notice that as the power $n$ gets larger, the curves tend to flatten toward the $x$-axis on the interval $(-1, 1)$, and to rise more steeply for $|x| > 1$. Each curve passes through the point $(1, 1)$ and through the origin. The graphs of functions with even powers are symmetric about the $y$-axis; those with odd powers are symmetric about the origin. The even-powered functions are decreasing on the interval $(-\infty, 0]$ and increasing on $[0, \infty)$; the odd-powered functions are increasing over the entire real line $(-\infty, \infty)$.

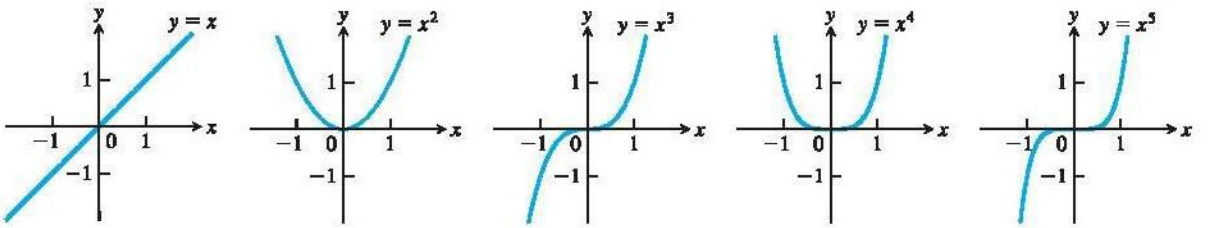



**FIGURE 1.15** Graphs of $f(x) = x^n$, $n = 1, 2, 3, 4, 5$, defined for $-\infty < x < \infty$.

**(b)** $a = -1$ or $a = -2$.

The graphs of the functions $f(x) = x^{-1} = 1/x$ and $g(x) = x^{-2} = 1/x^2$ are shown in Figure 1.16. Both functions are defined for all $x \neq 0$ (you can never divide by zero). The graph of $y = 1/x$ is the hyperbola $xy = 1$, which approaches the coordinate axes far from the origin. The graph of $y = 1/x^2$ also approaches the coordinate axes. The graph of the function $f$ is symmetric about the origin; $f$ is decreasing on the intervals $(-\infty, 0)$ and $(0, \infty)$. The graph of the function $g$ is symmetric about the $y$-axis; $g$ is increasing on $(-\infty, 0)$ and decreasing on $(0, \infty)$.

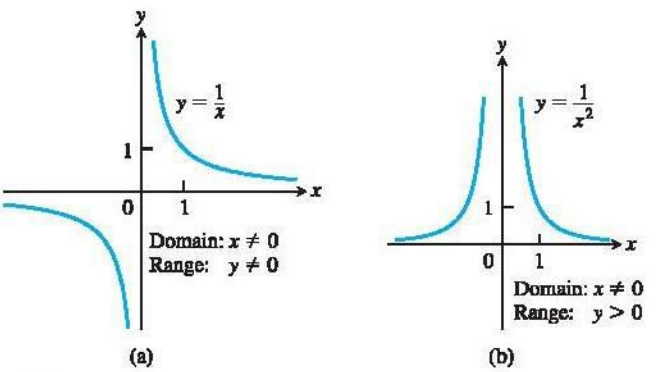



**FIGURE 1.16** Graphs of the power functions $f(x) = x^a$ for part (a) $a = -1$ and for part (b) $a = -2$.

**(c)** $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}$, and $\frac{2}{3}$.

The functions $f(x) = x^{1/2} = \sqrt{x}$ and $g(x) = x^{1/3} = \sqrt[3]{x}$ are the **square root** and **cube root** functions, respectively. The domain of the square root function is $[0, \infty)$, but the cube root function is defined for all real $x$. Their graphs are displayed in Figure 1.17, along with the graphs of $y = x^{3/2}$ and $y = x^{2/3}$. (Recall that $x^{3/2} = (x^{1/2})^3$ and $x^{2/3} = (x^{1/3})^2$.)

**Polynomials** A function $p$ is a **polynomial** if

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where $n$ is a nonnegative integer and the numbers $a_0, a_1, a_2, \ldots, a_n$ are real constants (called the **coefficients** of the polynomial). All polynomials have domain $(-\infty, \infty)$. If the

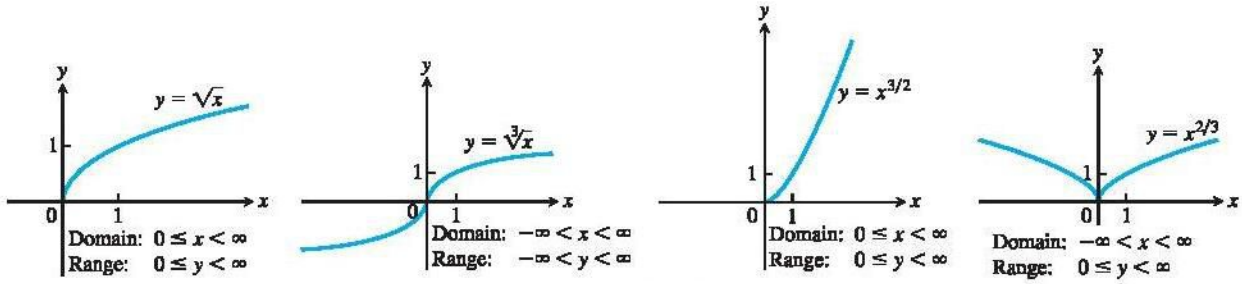



**FIGURE 1.17** Graphs of the power functions $f(x) = x^a$ for $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}$, and $\frac{2}{3}$.

leading coefficient $a_n \neq 0$ and $n > 0$, then $n$ is called the **degree** of the polynomial. Linear functions with $m \neq 0$ are polynomials of degree 1. Polynomials of degree 2, usually written as $p(x) = ax^2 + bx + c$, are called **quadratic functions**. Likewise, **cubic functions** are polynomials $p(x) = ax^3 + bx^2 + cx + d$ of degree 3. Figure 1.18 shows the graphs of three polynomials. Techniques to graph polynomials are studied in Chapter 4.

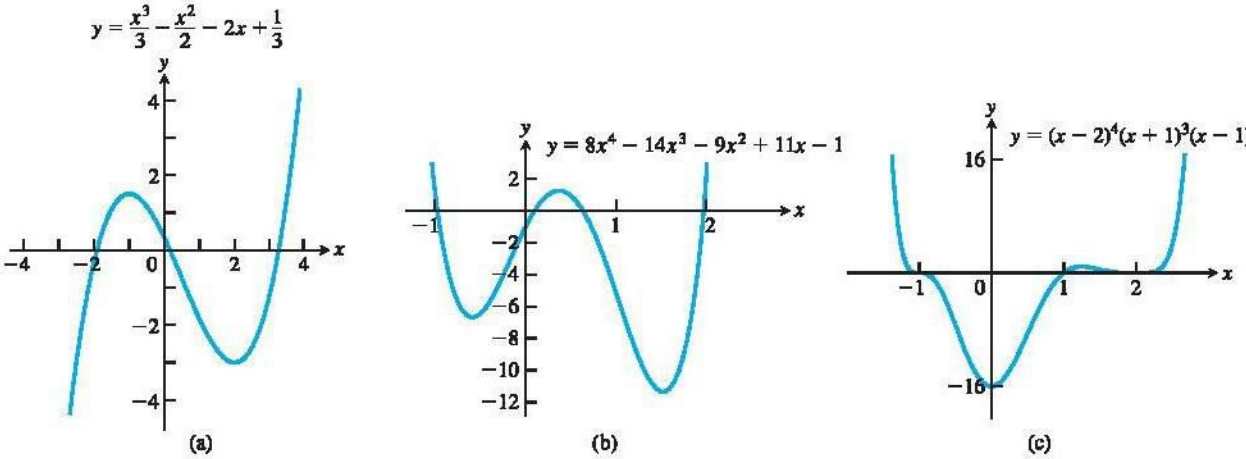



**FIGURE 1.18** Graphs of three polynomial functions.

**Rational Functions** A **rational function** is a quotient or ratio $f(x) = p(x)/q(x)$, where $p$ and $q$ are polynomials. The domain of a rational function is the set of all real $x$ for which $q(x) \neq 0$. The graphs of several rational functions are shown in Figure 1.19.

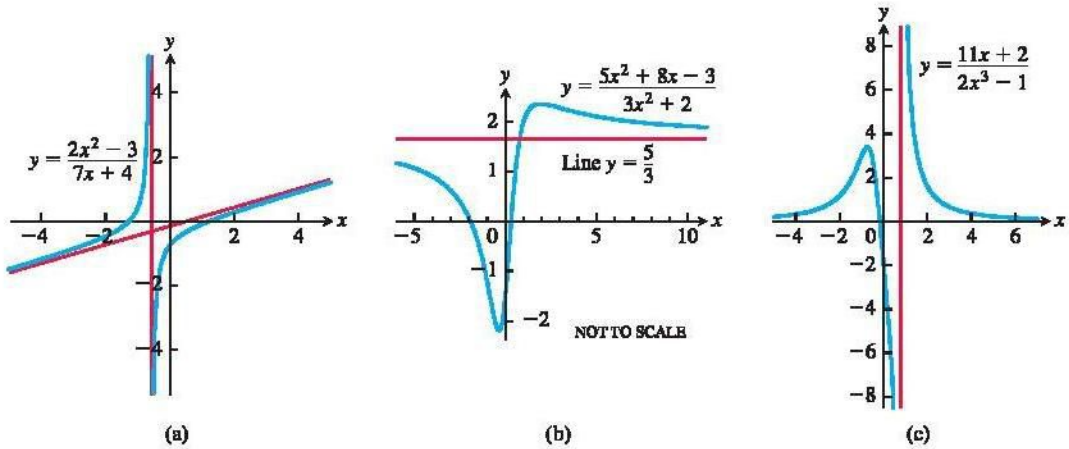



**FIGURE 1.19** Graphs of three rational functions. The straight red lines approached by the graphs are called *asymptotes* and are not part of the graphs. We discuss asymptotes in Section 2.6.

**Algebraic Functions** Any function constructed from polynomials using algebraic operations (addition, subtraction, multiplication, division, and taking roots) lies within the class of **algebraic functions**. All rational functions are algebraic, but also included are more complicated functions (such as those satisfying an equation like $y^3 - 9xy + x^3 = 0$, studied in Section 3.7). Figure 1.20 displays the graphs of three algebraic functions.

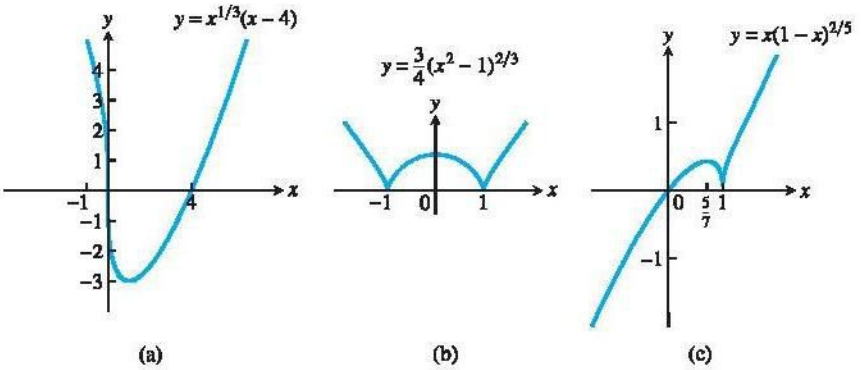



**FIGURE 1.20** Graphs of three algebraic functions.

**Trigonometric Functions** The six basic trigonometric functions are reviewed in Section 1.3. The graphs of the sine and cosine functions are shown in Figure 1.21.

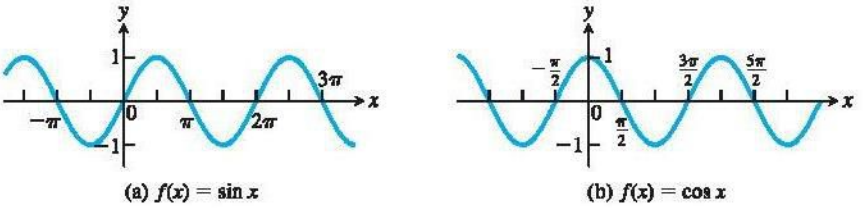



**FIGURE 1.21** Graphs of the sine and cosine functions.

**Exponential Functions** Functions of the form $f(x) = a^x$, where the base $a > 0$ is a positive constant and $a \neq 1$, are called **exponential functions**. All exponential functions have domain $(-\infty, \infty)$ and range $(0, \infty)$, so an exponential function never assumes the value 0. We develop exponential functions in Section 7.3. The graphs of some exponential functions are shown in Figure 1.22.

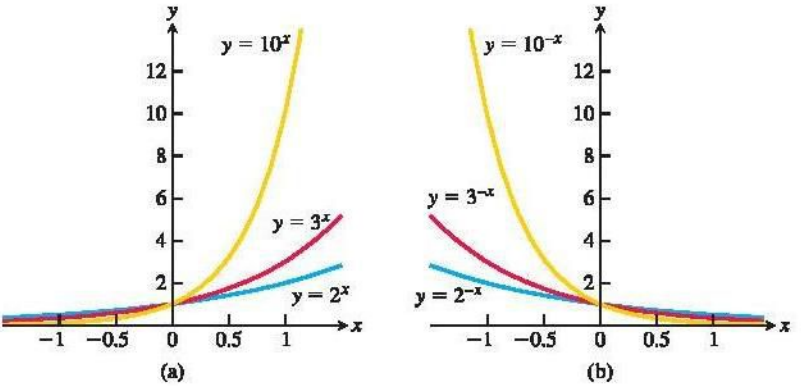



**FIGURE 1.22** Graphs of exponential functions.

**Logarithmic Functions** These are the functions $f(x) = \log_a x$, where the base $a \neq 1$ is a positive constant. They are the *inverse functions* of the exponential functions, and we define these functions in Section 7.2. Figure 1.23 shows the graphs of four logarithmic functions with various bases. In each case the domain is $(0, \infty)$ and the range is $(-\infty, \infty)$.

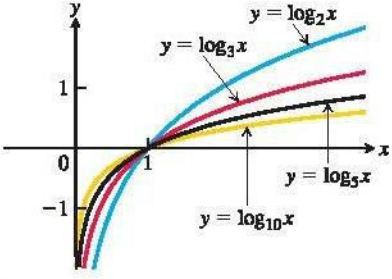



**FIGURE 1.23** Graphs of four logarithmic functions.

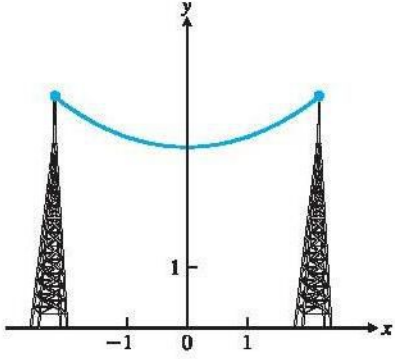



**FIGURE 1.24** Graph of a catenary or hanging cable. (The Latin word *catena* means "chain.")

**Transcendental Functions** These are functions that are not algebraic. They include the trigonometric, inverse trigonometric, exponential, and logarithmic functions, and many other functions as well. A particular example of a transcendental function is a **catenary**. Its graph has the shape of a cable, like a telephone line or electric cable, strung from one support to another and hanging freely under its own weight (Figure 1.24). The function defining the graph is discussed in Section 7.7.

# 1.2 Combining Functions; Shifting and Scaling Graphs

In this section we look at the main ways functions are combined or transformed to form new functions.

## Sums, Differences, Products, and Quotients

Like numbers, functions can be added, subtracted, multiplied, and divided (except where the denominator is zero) to produce new functions. If $f$ and $g$ are functions, then for every $x$ that belongs to the domains of both $f$ and $g$ (that is, for $x \in D(f) \cap D(g)$), we define functions $f + g$, $f - g$, and $fg$ by the formulas

$$(f + g)(x) = f(x) + g(x)$$
$$(f - g)(x) = f(x) - g(x)$$
$$(fg)(x) = f(x)g(x).$$

Notice that the + sign on the left-hand side of the first equation represents the operation of addition of *functions*, whereas the + on the right-hand side of the equation means addition of the real numbers $f(x)$ and $g(x)$.

At any point of $D(f) \cap D(g)$ at which $g(x) \neq 0$, we can also define the function $f/g$ by the formula

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \qquad (\text{where } g(x) \neq 0).$$

Functions can also be multiplied by constants: If $c$ is a real number, then the function $cf$ is defined for all $x$ in the domain of $f$ by

$$(cf)(x) = cf(x).$$

**EXAMPLE 1** The functions defined by the formulas

$$f(x) = \sqrt{x} \qquad \text{and} \qquad g(x) = \sqrt{1 - x}$$

have domains $D(f) = [0, \infty)$ and $D(g) = (-\infty, 1]$. The points common to these domains are the points

$$[0, \infty) \cap (-\infty, 1] = [0, 1].$$

The following table summarizes the formulas and domains for the various algebraic combinations of the two functions. We also write $f \cdot g$ for the product function $fg$.

| Function | Formula | Domain |
|---|---|---|
| $f + g$ | $(f + g)(x) = \sqrt{x} + \sqrt{1 - x}$ | $[0, 1] = D(f) \cap D(g)$ |
| $f - g$ | $(f - g)(x) = \sqrt{x} - \sqrt{1 - x}$ | $[0, 1]$ |
| $g - f$ | $(g - f)(x) = \sqrt{1 - x} - \sqrt{x}$ | $[0, 1]$ |
| $f \cdot g$ | $(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1 - x)}$ | $[0, 1]$ |
| $f/g$ | $\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \sqrt{\frac{x}{1 - x}}$ | $[0, 1)$ ($x = 1$ excluded) |
| $g/f$ | $\frac{g}{f}(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1 - x}{x}}$ | $(0, 1]$ ($x = 0$ excluded) |

■

The graph of the function $f + g$ is obtained from the graphs of $f$ and $g$ by adding the corresponding $y$-coordinates $f(x)$ and $g(x)$ at each point $x \in D(f) \cap D(g)$, as in Figure 1.25. The graphs of $f + g$ and $f \cdot g$ from Example 1 are shown in Figure 1.26.

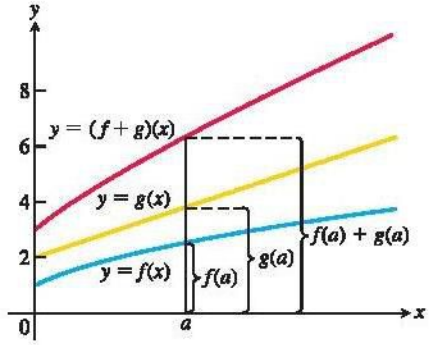



**FIGURE 1.25** Graphical addition of two functions.

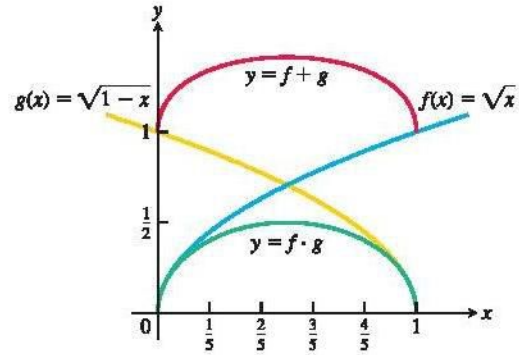



**FIGURE 1.26** The domain of the function $f + g$ is the intersection of the domains of $f$ and $g$, the interval $[0, 1]$ on the $x$-axis where these domains overlap. This interval is also the domain of the function $f \cdot g$ (Example 1).

## Composite Functions

Composition is another method for combining functions.

> **DEFINITION** If $f$ and $g$ are functions, the **composite** function $f \circ g$ ("$f$ composed with $g$") is defined by
>
> $$(f \circ g)(x) = f(g(x)).$$
>
> The domain of $f \circ g$ consists of the numbers $x$ in the domain of $g$ for which $g(x)$ lies in the domain of $f$.

The definition implies that $f \circ g$ can be formed when the range of $g$ lies in the domain of $f$. To find $(f \circ g)(x)$, *first* find $g(x)$ and *second* find $f(g(x))$. Figure 1.27 pictures $f \circ g$ as a machine diagram, and Figure 1.28 shows the composite as an arrow diagram.

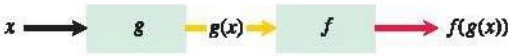



**FIGURE 1.27** A composite function $f \circ g$ uses the output $g(x)$ of the first function $g$ as the input for the second function $f$.

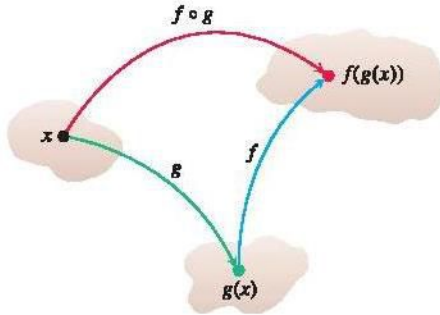



**FIGURE 1.28** Arrow diagram for $f \circ g$. If $x$ lies in the domain of $g$ and $g(x)$ lies in the domain of $f$, then the functions $f$ and $g$ can be composed to form $(f \circ g)(x)$.

To evaluate the composite function $g \circ f$ (when defined), we find $f(x)$ first and then $g(f(x))$. The domain of $g \circ f$ is the set of numbers $x$ in the domain of $f$ such that $f(x)$ lies in the domain of $g$.

The functions $f \circ g$ and $g \circ f$ are usually quite different.

**EXAMPLE 2** If $f(x) = \sqrt{x}$ and $g(x) = x + 1$, find

**(a)** $(f \circ g)(x)$ **(b)** $(g \circ f)(x)$ **(c)** $(f \circ f)(x)$ **(d)** $(g \circ g)(x)$.

**Solution**

| | Composite | Domain |
|---|---|---|
| **(a)** | $(f \circ g)(x) = f(g(x)) = \sqrt{g(x)} = \sqrt{x+1}$ | $[-1, \infty)$ |
| **(b)** | $(g \circ f)(x) = g(f(x)) = f(x) + 1 = \sqrt{x} + 1$ | $[0, \infty)$ |
| **(c)** | $(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{1/4}$ | $[0, \infty)$ |
| **(d)** | $(g \circ g)(x) = g(g(x)) = g(x) + 1 = (x + 1) + 1 = x + 2$ | $(-\infty, \infty)$ |

To see why the domain of $f \circ g$ is $[-1, \infty)$, notice that $g(x) = x + 1$ is defined for all real $x$ but belongs to the domain of $f$ only if $x + 1 \geq 0$, that is to say, when $x \geq -1$. ■

Notice that if $f(x) = x^2$ and $g(x) = \sqrt{x}$, then $(f \circ g)(x) = \left(\sqrt{x}\right)^2 = x$. However, the domain of $f \circ g$ is $[0, \infty)$, not $(-\infty, \infty)$, since $\sqrt{x}$ requires $x \geq 0$.

## Shifting a Graph of a Function

A common way to obtain a new function from an existing one is by adding a constant to each output of the existing function, or to its input variable. The graph of the new function is the graph of the original function shifted vertically or horizontally, as follows.

**Shift Formulas**

**Vertical Shifts**

| | |
|---|---|
| $y = f(x) + k$ | Shifts the graph of $f$ *up* $k$ units if $k > 0$<br>Shifts it *down* $\lvert k \rvert$ units if $k < 0$ |

**Horizontal Shifts**

| | |
|---|---|
| $y = f(x + h)$ | Shifts the graph of $f$ *left* $h$ units if $h > 0$<br>Shifts it *right* $\lvert h \rvert$ units if $h < 0$ |

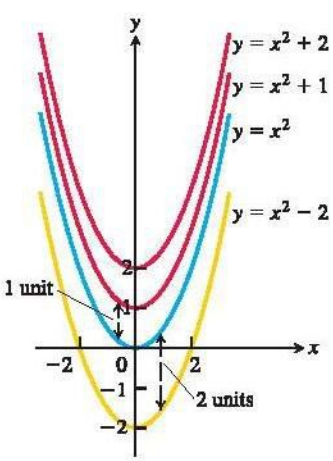



**FIGURE 1.29** To shift the graph of $f(x) = x^2$ up (or down), we add positive (or negative) constants to the formula for $f$ (Examples 3a and b).

**EXAMPLE 3**

**(a)** Adding 1 to the right-hand side of the formula $y = x^2$ to get $y = x^2 + 1$ shifts the graph up 1 unit (Figure 1.29).

**(b)** Adding $-2$ to the right-hand side of the formula $y = x^2$ to get $y = x^2 - 2$ shifts the graph down 2 units (Figure 1.29).

**(c)** Adding 3 to $x$ in $y = x^2$ to get $y = (x + 3)^2$ shifts the graph 3 units to the left, while adding $-2$ shifts the graph 2 units to the right (Figure 1.30).

**(d)** Adding $-2$ to $x$ in $y = |x|$, and then adding $-1$ to the result, gives $y = |x - 2| - 1$ and shifts the graph 2 units to the right and 1 unit down (Figure 1.31). ■

## Scaling and Reflecting a Graph of a Function

To scale the graph of a function $y = f(x)$ is to stretch or compress it, vertically or horizontally. This is accomplished by multiplying the function $f$, or the independent variable $x$, by an appropriate constant $c$. Reflections across the coordinate axes are special cases where $c = -1$.

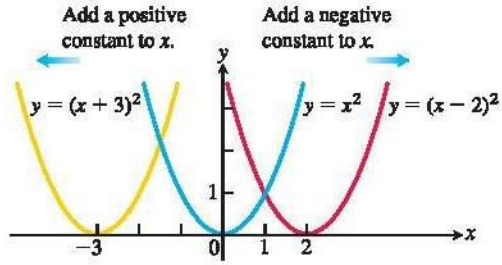



**FIGURE 1.30** To shift the graph of $y = x^2$ to the left, we add a positive constant to $x$ (Example 3c). To shift the graph to the right, we add a negative constant to $x$.

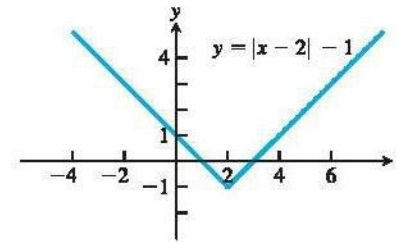



**FIGURE 1.31** The graph of $y = |x|$ shifted 2 units to the right and 1 unit down (Example 3d).

### Vertical and Horizontal Scaling and Reflecting Formulas

**For $c > 1$, the graph is scaled:**

| | |
|---|---|
| $y = cf(x)$ | Stretches the graph of $f$ vertically by a factor of $c$. |
| $y = \frac{1}{c}f(x)$ | Compresses the graph of $f$ vertically by a factor of $c$. |
| $y = f(cx)$ | Compresses the graph of $f$ horizontally by a factor of $c$. |
| $y = f(x/c)$ | Stretches the graph of $f$ horizontally by a factor of $c$. |

**For $c = -1$, the graph is reflected:**

| | |
|---|---|
| $y = -f(x)$ | Reflects the graph of $f$ across the $x$-axis. |
| $y = f(-x)$ | Reflects the graph of $f$ across the $y$-axis. |

**EXAMPLE 4** Here we scale and reflect the graph of $y = \sqrt{x}$.

(a) **Vertical:** Multiplying the right-hand side of $y = \sqrt{x}$ by 3 to get $y = 3\sqrt{x}$ stretches the graph vertically by a factor of 3, whereas multiplying by $1/3$ compresses the graph by a factor of 3 (Figure 1.32).

(b) **Horizontal:** The graph of $y = \sqrt{3x}$ is a horizontal compression of the graph of $y = \sqrt{x}$ by a factor of 3, and $y = \sqrt{x/3}$ is a horizontal stretching by a factor of 3 (Figure 1.33). Note that $y = \sqrt{3x} = \sqrt{3}\sqrt{x}$ so a horizontal compression *may* correspond to a vertical stretching by a different scaling factor. Likewise, a horizontal stretching may correspond to a vertical compression by a different scaling factor.

(c) **Reflection:** The graph of $y = -\sqrt{x}$ is a reflection of $y = \sqrt{x}$ across the $x$-axis, and $y = \sqrt{-x}$ is a reflection across the $y$-axis (Figure 1.34). ■

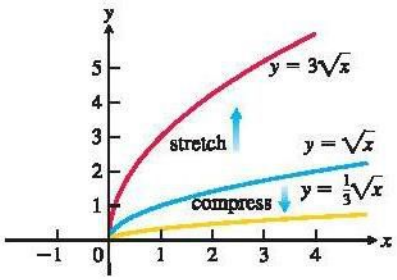



**FIGURE 1.32** Vertically stretching and compressing the graph $y = \sqrt{x}$ by a factor of 3 (Example 4a).

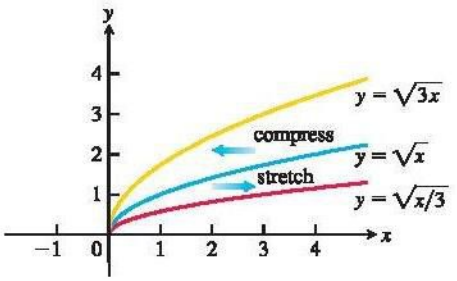



**FIGURE 1.33** Horizontally stretching and compressing the graph $y = \sqrt{x}$ by a factor of 3 (Example 4b).

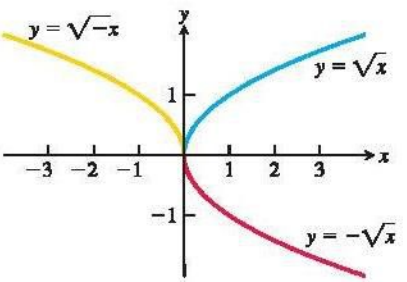



**FIGURE 1.34** Reflections of the graph $y = \sqrt{x}$ across the coordinate axes (Example 4c).

**EXAMPLE 5** Given the function $f(x) = x^4 - 4x^3 + 10$ (Figure 1.35a), find formulas to

(a) compress the graph horizontally by a factor of 2 followed by a reflection across the $y$-axis (Figure 1.35b).

(b) compress the graph vertically by a factor of 2 followed by a reflection across the $x$-axis (Figure 1.35c).

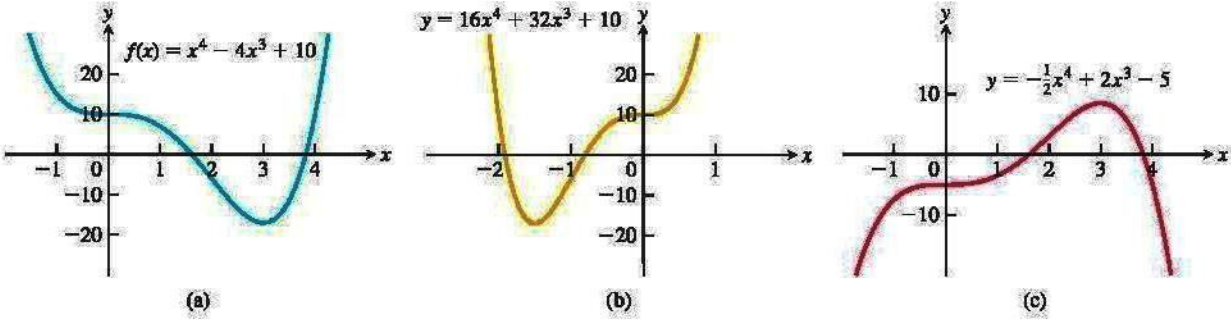



**FIGURE 1.35** (a) The original graph of $f$. (b) The horizontal compression of $y = f(x)$ in part (a) by a factor of 2, followed by a reflection across the $y$-axis. (c) The vertical compression of $y = f(x)$ in part (a) by a factor of 2, followed by a reflection across the $x$-axis (Example 5).

**Solution**

(a) We multiply $x$ by 2 to get the horizontal compression, and by $-1$ to give reflection across the $y$-axis. The formula is obtained by substituting $-2x$ for $x$ in the right-hand side of the equation for $f$:

$$\begin{aligned} y = f(-2x) &= (-2x)^4 - 4(-2x)^3 + 10 \\ &= 16x^4 + 32x^3 + 10. \end{aligned}$$

(b) The formula is

$$y = -\frac{1}{2}f(x) = -\frac{1}{2}x^4 + 2x^3 - 5.$$

■